MD. RESHAD AL MUTTAKI

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SUMMARY

EEE graduate from BRAC University with a research focus on semiconductor device design and simulation, seeking a Lecturer position in the Department of Computer Science and Engineering (CSE) at BRAC University. Experienced in TCAD-based device modeling, VLSI design, digital-twin-driven optimization, and undergraduate teaching and mentorship, with six peer-reviewed Q1 journal publications and multiple manuscripts under review and expected to be published very soon. Former Student Tutor and Student Mentor in the Department of EEE, with a background spanning VLSI and mixed-signal design, committed to a career in academia focused on teaching, research, and innovation.

EDUCATION

Bachelor of Science in Electrical and Electronic Engineering — *Jan 2022 – Jan 2026*
BRAC University — Major: Electronics
Relevant coursework in Electrical and Electronic Engineering — **CGPA: 3.81**

Higher Secondary Certificate (HSC) — *Passing Year: 2020*
Square School & College | Rajshahi Board — Major: Science **GPA: 5.00**

Secondary School Certificate (SSC) — *Passing Year: 2018*
Square School & College | Rajshahi Board — Major: Science **GPA: 5.00**

PROFESSIONAL EXPERIENCE

Research Fellow — *May 2026*
BRAC University
- Conducting research on "GaN-Based G4FET Technology: Fabrication, Characterization, Modeling, and Circuit Demonstration," funded under the CREST initiative.
- Designing a 5 nm AlGaN/GaN tri-gate FinFET (AS^3-FinFET) in Silvaco Atlas TCAD, applying VLSI short-channel-effect theory and polarization-charge device physics.
- Applying gate, doping, and spacer engineering principles to manage electrostatic scaling and interface effects at the sub-5 nm VLSI node.
- Benchmarking the design against global FinFET/nanosheet device approaches, with a roadmap toward compact modeling and circuit-level demonstration.

Research Assistant (RA) — *Mar 2026*
BRAC University
- Conducting research on "Advancing Sustainable Biosensing: A Self-powered Biosensor for Real-time and Wireless Health Monitoring," funded by RSGI of BRAC University.
- Developing an e-skin-type biosensor in both simulation and hardware environments.

Student Tutor (ST), BSRM SoE, Dept. of EEE — *Jan 2024 – Oct 2025*
BRAC University
- Managed and mentored undergraduate students through tutorial and consultation sessions in both online and offline environments across multiple courses.
- Prepared supplementary problem sets, worked examples, and exam-review materials to reinforce lecture content and address common conceptual gaps.
- Held regular office hours and pre-exam review sessions, helping students build stronger fundamentals in circuit analysis, semiconductor devices, and energy-conversion topics.

Student Mentor, FYAT, OAA — *Jan 2024 – Jan 2025*
BRAC University
- Provided one-on-one mentorship to students as part of the First Year Advising Team (FYAT).
- Guided first-year students on course selection, study habits, and adjustment to university life.

COURSES TAUGHT AS STUDENT TUTOR (ST)

Electrical Circuits I (EEE/ECE101) *Summer 2024*

- Foundational DC circuit analysis: Kirchhoff's laws, nodal/mesh analysis, Thevenin, Norton and superposition theorems, and transient response of RC/RL circuits.

Electronic Circuits I (EEE/ECE205) *Spring 2024*

- Semiconductor device fundamentals and single-stage amplifier design covering diode, BJT, JFET, and MOSFET construction, characteristics, and circuit analysis.

Energy Conversion (EEE221) *Fall 2024 – Summer 2025*

- Principles, construction, and performance analysis of DC and AC energy-conversion machines, including transformers, induction motors, and synchronous machines.

RESEARCH INTERESTS

1. Electronics & Semiconductor

- VLSI Design
- Semiconductor Devices
- Embedded Systems
- CAD Algorithms

2. AI & Cybernetics

- Artificial Intelligence & ML
- IoT & Cyber-Physical Systems
- Digital Twin Systems
- Computer Architecture

3. Biomedical Engineering

- Healthcare Technology
- Biomedical AI
- Flexible PCB & E-Skin
- Health & Bioinformatics

ACHIEVEMENTS AND HONORS

- **Vice Chancellor's (VC's) List** — Spring 2023; Summer 2023; Summer 2024
- **Dean's List** — Fall 2022; Fall 2023; Spring 2024; Fall 2024; Spring 2025

PUBLICATIONS

1. **HERA: A Hierarchical Expected-Hypervolume Reinforcement-Guided Acquisition Framework and Digital Twin for Multi-Objective Optimization of Sub-3 nm Gate-All-Around FinFETs**
   Authors: **Md. Reshad Al Muttaki**, Tahmid Faiz Noor, SK Tahmed Salim Rafid
   Status: **Under Review** in *Materials Today Electronics* | **1st Author**
   A constrained multi-objective VLSI device-optimization framework for sub-3 nm gate-all-around FinFETs, developed to navigate the electrostatic and transport trade-offs inherent to aggressive gate-length scaling. The framework combines a Gaussian-process ensemble surrogate, a constrained expected-hypervolume acquisition function, and a reinforcement-learning-warm-started search to efficiently explore the device design space over a calibrated TCAD digital twin, with representative device designs cross-verified using DEVSIM drift-diffusion TCAD.

2. **Towards smarter grids: A systematic review of wide area monitoring enhancing stability, protection, and promoting environmental sustainability**
   Authors: **Md. Reshad Al Muttaki**, Alvi Ibn Amzad Anil, Sadia Afrin, Shameem Hasan
   Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**
   DOI: `10.1016/j.ecmx.2025.101424`

3. **AI-powered cybersecurity for smart grid communication: A systematic review of intrusion detection and threat mitigation systems**
   Authors: Sadia Afrin, **Md. Reshad Al Muttaki**, Alvi Ibn Amzad Anil, Shameem Hasan
   Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **2nd Author**
   DOI: `10.1016/j.ecmx.2025.101416`

4. **State-of-charge-aware charging opportunity detection for electric vehicles using data-driven learning and digital twin simulation**
   Authors: **Md. Reshad Al Muttaki**, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan
   Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **1st Author**
   DOI: `10.1016/j.ecmx.2026.101775`

5. **The next generation of energy storage for smart and sustainable power grids: A systematic review**
   Authors: Alvi Ibn Amzad Anil, **Md. Reshad Al Muttaki**, Sadia Afrin, Shameem Hasan
   Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **2nd Author**
   DOI: `10.1016/j.ecmx.2026.101771`

6. **Demand-based anomaly detection algorithm in electric vehicle charging systems using machine learning and digital twin simulation**
   Authors: **Md. Reshad Al Muttaki**, Sadia Afrin, Alvi Ibn Amzad Anil, Shameem Hasan
   Journal of Energy Storage | Q1, Top 6% | IF: 9.8 | CiteScore: 13.3 | **1st Author**
   DOI: `10.1016/j.est.2026.122795`

7. **Balancing cybersecurity, wireless charging, and safety: Digital twin-driven techno-economic analysis of electric vehicles**
   Authors: Shameem Hasan, Abu M. Fuad, B.M. Khalid Hasan Drobo, **Md. Reshad Al Muttaki**, Sadia Afrin, Alvi Ibn Amzad Anil, Omar Farrok
   Energy Conversion and Management: X | Q1, Top 7% | IF: 8.78 | CiteScore: 11.3 | **4th Author**
   DOI: `10.1016/j.ecmx.2026.102107`

CERTIFICATIONS

Industrial Training in VLSI Design *Oct 2025*
Ulkasemi Pvt. Ltd.

- Analog Design — circuit-level design and analysis of analog building blocks.
- Digital Design — RTL-level design methodology for digital systems.
- Digital Verification — functional verification flows for digital circuit designs.
- Custom Layout — physical layout design for custom VLSI cells.
- Digital Circuit Design — design and implementation of digital circuit blocks.

SKILLS

Programming:	Python, C, MATLAB, Verilog, VHDL
EDA & Simulation:	Quartus, DEVSIM, Cadence, COMSOL Multiphysics, Silvaco TCAD, Altium Designer
Web & Markup:	HTML, CSS, LaTeX
Tools & Platforms:	Git & GitHub, Google Suite, MS Office Suite

REFERENCES

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